

CREDIT SCORING MODEL

Predicting Individual Creditworthiness Using Machine Learning

Domain	Financial Analytics / Machine Learning
Techniques	Logistic Regression, Decision Trees, Random Forest
Task Type	Binary Classification (Creditworthy vs. Risky)
Key Metrics	Precision, Recall, F1-Score, ROC-AUC

Project Report

1. Project Objective

The objective of this project is to build a machine learning model that predicts an individual's **creditworthiness** — that is, the likelihood that a borrower will repay a loan or credit obligation responsibly — using their past financial history. The model outputs a classification (e.g., "Good Credit" vs. "Bad Credit") or a probability score that can be used by financial institutions to support lending decisions, set interest rates, and manage credit risk.

Traditional credit scoring relies heavily on manual rules and bureau scores such as FICO/CIBIL. This project aims to demonstrate how supervised machine learning classification algorithms can learn patterns from historical financial data to make accurate, data-driven, and scalable creditworthiness predictions.

2. Dataset Description

The model is trained on structured historical financial data. Publicly available datasets such as the **German Credit Data (UCI)**, **Lending Club Loan Dataset**, or **Home Credit Default Risk (Kaggle)** can be used, or a custom dataset can be compiled from banking/loan records. A typical dataset includes:

Feature Category	Example Fields
Demographic	Age, employment status, marital status, residence type
Income	Monthly/annual income, income source, employment tenure
Debt & Liabilities	Existing loans, credit card balances, debt-to-income ratio
Payment History	Past delinquencies, missed payments, days-past-due history
Credit Behaviour	Number of credit lines, credit utilization, credit age
Loan Details	Loan amount, purpose, tenure, interest rate
Target Variable	Credit risk label — "Good" / "Bad" (or Default: Yes/No)

Table 1: Representative feature categories in a credit scoring dataset.

3. Project Workflow

- **Data Collection** — Gather historical loan/credit records with a known outcome (default or not).
- **Data Cleaning** — Handle missing values, remove duplicates, treat outliers in income/debt fields.
- **Feature Engineering** — Derive new, more predictive variables from raw financial data.
- **Data Splitting** — Split into training and test sets (e.g., 80/20 or 70/30), often with stratification.
- **Model Building** — Train Logistic Regression, Decision Tree, and Random Forest classifiers.
- **Model Evaluation** — Assess performance using Precision, Recall, F1-Score, and ROC-AUC.
- **Model Selection & Tuning** — Compare models, tune hyperparameters, select the best performer.
- **Deployment (optional)** — Expose the model via an API/dashboard for real-time scoring.

4. Feature Engineering

Feature engineering is central to a good credit scoring model, since raw financial fields are often less predictive than derived ratios and behavioural indicators. Key engineered features include:

- **Debt-to-Income Ratio (DTI)** = Total Monthly Debt Payments / Monthly Income — a strong risk indicator.
- **Credit Utilization Ratio** = Outstanding Credit Balance / Total Credit Limit.
- **Payment Delinquency Score** — weighted count of late/missed payments in the last 12–24 months.
- **Credit History Length** — number of years since the first credit account was opened.
- **Number of Recent Credit Inquiries** — frequent inquiries can indicate financial distress.
- **Loan-to-Income Ratio** — requested loan amount relative to annual income.
- **Income Stability** — variance in income over recent periods, employment tenure.
- **Categorical Encoding** — one-hot/target encoding for employment type, loan purpose, residence type.
- **Binning** — converting continuous variables (e.g., age, income) into risk bands where useful.

Proper scaling (standardization/normalization) is applied for algorithms like Logistic Regression, while tree-based models (Decision Tree, Random Forest) are less sensitive to feature scale but benefit from careful handling of categorical variables and missing data imputation.

5. Modelling Approach

This project frames creditworthiness prediction as a **binary classification problem**. Three complementary algorithms are used to balance interpretability and predictive power:

Algorithm	Strengths	Considerations
Logistic Regression	Highly interpretable; coefficients show feature impact.	Assumes linear relationships; sensitive to feature scaling and multicollinearity.
Decision Tree	Captures non-linear relationships; easy to visualize.	Prone to overfitting; sensitive to small changes in data.
Random Forest	Ensemble of trees reduces overfitting; high accuracy.	Less interpretable than single trees; computationally intensive.

Table 2: Comparison of candidate classification algorithms.

5.1 Hyperparameter Tuning

Grid Search or Randomized Search with k-fold cross-validation is used to tune key hyperparameters — regularization strength (C) for Logistic Regression; max depth, min samples split for Decision Tree; and number of estimators, max depth, and max features for Random Forest — to prevent overfitting and improve generalization on unseen applicants.

6. Model Evaluation Metrics

Since credit datasets are typically **imbalanced** (far fewer defaulters than good borrowers), accuracy alone is a poor measure of performance. This project therefore emphasizes the following metrics:

Metric	What It Measures	Why It Matters for Credit Scoring
Precision	Of all applicants predicted as "good credit", how many actually were.	High precision minimizes approving loans that later default.
Recall (Sensitivity)	Of all actual good-credit applicants, how many were correctly identified.	High recall ensures not rejecting creditworthy customers.
F1-Score	Harmonic mean of Precision and Recall.	Balances false approvals against false rejections in a single score.
ROC-AUC	Model's ability to distinguish between good and bad credit applicants.	Widely used industry benchmark; threshold-independent measure.
Confusion Matrix	Breakdown of True/False Positives and Negatives.	Reveals exact error types — critical for risk-based decision-making.

Table 3: Evaluation metrics and their business relevance.

Additional techniques such as the **KS statistic** and **Gini coefficient** (common in credit risk industry practice) can also be reported alongside ROC-AUC for a more domain-specific assessment. Class imbalance is addressed using techniques such as SMOTE (oversampling), class-weighting, or undersampling of the majority class.

7. Tools & Technologies

Category	Tools/Libraries
Programming Language	Python
Data Handling	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Modelling	Scikit-learn (Logistic Regression, Decision Tree, Random Forest)
Imbalance Handling	imbalanced-learn (SMOTE)
Model Evaluation	Scikit-learn metrics module (precision_recall_fscore, roc_auc_score)
Deployment (optional)	Flask/FastAPI, Streamlit, Docker

8. Expected Outcomes

- A trained classification model capable of predicting creditworthiness with strong ROC-AUC performance.
- A ranked comparison of Logistic Regression, Decision Tree, and Random Forest on the same dataset.
- Feature importance insights identifying the strongest predictors of credit risk (e.g., DTI ratio, payment history).
- A reusable, documented pipeline (data cleaning → feature engineering → model training → evaluation) that can be extended to new financial datasets.

9. Limitations & Ethical Considerations

- **Bias & Fairness** — Historical data may encode societal biases; care must be taken to avoid discriminatory outcomes across protected attributes (e.g., gender, ethnicity).
- **Data Privacy** — Financial data is sensitive and must be handled in compliance with data protection regulations.
- **Model Interpretability** — Regulated lending decisions often require explainable models; techniques like SHAP values can be used to interpret Random Forest predictions.
- **Concept Drift** — Economic conditions change over time, so models require periodic retraining and monitoring.

10. Conclusion

This project demonstrates an end-to-end approach to credit scoring using classical machine learning classification techniques. By combining thoughtful feature engineering from financial history with robust model evaluation using Precision, Recall, F1-Score, and ROC-AUC, the resulting system offers a transparent, data-driven alternative to manual credit assessment — helping financial institutions make faster, fairer, and more accurate lending decisions.